AI-BASED VOICE COMPLAINT MANAGEMENT SYSTEM

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfillment for the award of the degree of

DSA0306 – Natural Language Processing

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Submitted by

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DECLARATION

We, **M JEEVITHA (192424257)** and of the Department of B. Tech Artificial Intelligence and Data Science (AI&DS), SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "**AI-BASED VOICE COMPLAINT MANAGEMENT SYSTEM USING NATURAL LANGUAGE PROCESSING**" is the result of our own Bonafide efforts.

To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged.

We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: 02/01/26

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "**AI-BASED VOICE COMPLAINT MANAGEMENT SYSTEM USING NATURAL LANGUAGE PROCESSING**" has been carried out by **M JEEVITHA (192424257)** under the supervision of **Dr KUMARAGURUBARAN T** and is submitted in partial fulfilment of the requirements for the B.Tech Artificial Intelligence and Data Science programme at Saveetha Institute of Medical and Technical Sciences, Chennai.

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Praharshini	Voice Input & Speech-to-Text	Developed voice input, audio recording and speech-to-text conversion using ASR	Tested voice recording, speech recognition and transcription accuracy under different conditions	Contributed to Introduction, Literature Review and Module 1 documentation	25%
Deeksha Sree	NLP Processing & Feature Extraction	Developed text normalization, tokenization, lemmatization and TF-IDF feature extraction	Tested NLP preprocessing and TF-IDF feature generation using different complaints	Contributed to Engineering Design and Module 2 documentation	25%
Hitaishi	AI Complaint Classification	Developed complaint classification using Multinomial Naive Bayes and implemented category/priority prediction	Tested classification accuracy, confidence scores and prediction of new complaints	Contributed to Implementation, Results and Module 3 documentation	25%
Jeevitha	Complaint Routing, Database & Dashboard	Developed department routing, database storage, complaint tracking, dashboard and report generation	Tested database operations, routing, dashboard functionality and complaint status updates	Contributed to Conclusion, Future Work and Module 4 documentation	25%
Total	—	—	—	—	100%

ABSTRACT

The AI-Based Voice Complaint Management System is proposed to simplify the process of registering, understanding, categorizing, prioritizing, tracking, and resolving complaints through spoken input. Conventional complaint-management systems often require users to type detailed descriptions, select categories manually, and repeatedly follow up to determine the status of a complaint. This can increase the effort required from users and may create difficulties for people who are less comfortable with text-based interfaces. The proposed system accepts a voice complaint, converts speech into text using an automatic speech recognition component, applies natural language processing to identify the complaint category and relevant information, assigns an appropriate priority, and stores the complaint in a structured database. A management dashboard can provide authorized personnel with complaint details, categories, priority levels, status, and resolution progress. The system can also support acknowledgement and status notifications. The engineering design emphasizes speech recognition quality, language and accent variation, complaint classification, privacy, security, usability, and reliable workflow management. The expected outcome is an accessible and efficient complaint-management platform that reduces manual effort and improves the traceability of complaint handling.

Keywords: voice complaint management; speech recognition; natural language processing; complaint classification; priority detection; AI; dashboard

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
NLP	Natural Language Processing	—
AI	Artificial Intelligence	—
STT	Speech-to-Text	—
TF-IDF	Term Frequency–Inverse Document Frequency	—
ML	Machine Learning	—
GUI	Graphical User Interface	—

CHAPTER 1
INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Complaint management is an important service activity in universities, public institutions, organizations, service centres, residential communities, and other environments where users need to report problems and receive appropriate responses. Traditional complaint systems commonly depend on written forms, web forms, emails, or manually recorded telephone calls. Although these mechanisms are useful, they can require considerable effort from the complainant and from staff responsible for converting unstructured complaints into actionable records.

Voice interfaces provide an alternative means of interaction. A user can describe a problem naturally instead of typing a long complaint. Artificial intelligence can then convert the spoken statement into text and analyse the content. Natural language processing can identify the probable category of the complaint, extract important keywords or entities, determine urgency where appropriate, and route the complaint to the responsible department. A centralized database and dashboard can maintain the complaint lifecycle from registration to closure.

The proposed project combines speech processing, NLP, classification, database management, and web-based visualization into one workflow. The system is intended as a practical engineering solution rather than a simple speech-to-text demonstration. Its usefulness depends on the accuracy of transcription, correct interpretation of complaints, appropriate prioritization, reliable storage, and clear presentation of complaint status.

1.2 Need for the Project

- Users may find typing detailed complaints inconvenient, particularly on mobile devices.
- Manual complaint registration can introduce delays and inconsistent categorization.
- Free-form voice complaints contain useful information that can be difficult to organize manually.
- Complaint handlers need a structured view of category, priority, status, assigned department, and resolution progress.

- Automated classification can reduce repetitive administrative work while retaining human oversight.
- A searchable complaint history can improve accountability and follow-up.
- A voice-first interface can improve accessibility and convenience for users.

1.3 Problem Statement

The engineering problem is to design and implement an AI-assisted complaint-management system that accepts spoken complaints, converts them into usable text, extracts and classifies complaint information, records the complaint in a structured database, assigns or recommends an appropriate priority and responsible category, and provides a dashboard for monitoring and resolution. The solution must account for speech-recognition errors, variation in accents and speaking styles, ambiguous language, privacy of complaint content, authentication, data security, system response time, and the need for human verification before consequential decisions are made.

1.4 Project Aim

To design and develop an AI-based voice complaint management system that converts spoken complaints into structured, searchable, prioritized, and trackable complaint records while reducing manual effort and improving the efficiency of complaint handling.

1.5 Project Objectives

To collect or create a suitable dataset of representative complaint utterances for system development and evaluation.

To capture user voice input and convert speech into text using an automatic speech recognition component.

To preprocess and normalize complaint text for subsequent NLP analysis.

To classify complaints into relevant categories such as infrastructure, maintenance, academic/service, transport, hostel, cleanliness, or other domain-specific categories.

To identify complaint priority or urgency using rule-based and/or machine-learning approaches.

To store complaint details securely in a structured database with unique complaint identification.

To provide an administrative dashboard for viewing, filtering, assigning, updating, and tracking complaints.

To evaluate the system using speech-recognition, classification, functional, integration, and usability measures.

1.6 Scope

Included in the project:

- Voice-based complaint capture through a microphone-enabled interface.
- Speech-to-text conversion and text preprocessing.
- NLP-based complaint category classification.
- Priority recommendation based on complaint content and defined rules/model output.
- Structured complaint storage and unique complaint ID generation.
- Complaint status management such as submitted, assigned, in progress, resolved, and closed.
- Authorized dashboard access for complaint monitoring and management.
- Testing using representative complaint samples and recorded test cases.

Not included in the project:

- Fully autonomous resolution of complaints without human intervention.
- Guaranteed recognition of every accent, language, background-noise condition, or speaking style.
- Automatic decisions involving disciplinary, legal, medical, financial, or safety consequences.
- Unrestricted access to complaint recordings or personal information.
- Large-scale production deployment unless separately implemented and validated.

Assumptions and boundary conditions: the prototype will use an appropriately consented or institutionally permitted dataset; complaint categories will be defined for the selected deployment domain; model performance will be reported using actual measured test results; and authorized human staff will remain responsible for final complaint assignment, priority confirmation, and resolution decisions.

1.7 Expected Engineering Outcome

The intended outcome is an integrated software platform consisting of voice capture, speech recognition, NLP-based complaint understanding, classification and priority logic, secure complaint storage, and a management dashboard. The system should demonstrate that voice input can be converted into structured complaint records and routed through a traceable workflow. The final report should present measured performance, limitations, testing evidence, and individual contributions rather than relying only on qualitative claims.

CHAPTER 2
LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review should examine research and established technologies related to automatic speech recognition, speech-to-text systems, natural language processing, text classification, intent detection, sentiment or urgency analysis, conversational interfaces, and automated complaint-management systems. For the final submitted report, the team should record the actual databases searched, search terms, publication period, inclusion and exclusion criteria, and the number of studies selected for detailed review.

2.2 Review of Existing Work

Existing complaint systems commonly provide web forms, mobile forms, email channels, call-centre registration, or ticketing interfaces. These systems can provide useful tracking and workflow management, but text-heavy interfaces may require users to manually formulate and categorize their complaints. A voice-enabled approach shifts part of this effort to speech recognition and NLP.

Automatic speech recognition converts spoken language into textual sequences. Its effectiveness can be influenced by acoustic conditions, pronunciation, accent, speaking rate, microphone quality, and domain vocabulary. NLP methods can subsequently transform the transcript into features or embeddings for classification. Traditional machine-learning models may use TF-IDF or related representations, while modern transformer-based models can capture richer contextual information.

Complaint classification can be formulated as an intent or topic classification problem. Priority estimation can combine explicit urgency words, complaint category, severity indicators, and organizational rules. However, automated priority decisions should be treated as recommendations when incorrect prioritization could affect users significantly. Human verification and an auditable workflow are therefore important design considerations.

2.3 Gap Identified

A useful gap for this project is the integration of multiple AI and software components into a single complaint lifecycle. Speech recognition alone does not provide complaint management, while a conventional ticketing system does not automatically interpret spoken language. The proposed system combines voice input, transcription, complaint understanding, category and priority recommendation, database storage, dashboard monitoring, and status tracking in one workflow.

2.4 Comparison of Existing Approaches

Table 2.1: Comparative analysis of existing complaint approaches

Approach	**Input**	**Classification**	**Tracking**	**Major Limitation**
Paper/manual complaint	Written	Manual	Limited	Slow and difficult to search
Web complaint form	Typed text	Manual/optional	Usually available	Typing and manual categorization
Call-centre registration	Voice	Manual	Available	Operator effort and inconsistency
Voice + ticketing	Voice	AI/Rules	Available	Depends on speech and NLP accuracy
Proposed system	Voice + text	AI-assisted	Dashboard	Requires reliable ASR, NLP and secure data handling

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Stakeholders and User Requirements

Table 3.1. Stakeholder requirements

Stakeholder	**Requirement**
Complainant	Submit complaint easily through voice or text and receive complaint ID/status.
Complaint staff	View, verify, categorize, assign, update and close complaints.
Department/administrator	Filter complaints by category, priority and status.
Management	Monitor complaint trends, response status and unresolved cases.
System administrator	Manage users, access rights, configuration and system availability.

3.2 Functional and Non-Functional Requirements

Type	**Requirement**
Functional	Capture voice input
Functional	Convert speech to text
Functional	Classify complaint category
Functional	Recommend priority
Functional	Generate complaint ID
Functional	Store and retrieve complaint records
Functional	Update complaint status and assignment
Functional	Display dashboard and search/filter results
Non-functional	Usable interface and clear feedback
Non-functional	Reasonable response time
Non-functional	Secure authentication and authorization
Non-functional	Reliable database storage
Non-functional	Maintainable modular architecture

3.3 Proposed System Architecture

The proposed architecture can be organized into six layers: (1) user interface and voice capture, (2) speech recognition, (3) NLP preprocessing and complaint understanding, (4) classification and priority recommendation, (5) complaint database and workflow management, and (6) administrator dashboard and notification services.

Typical flow: User speaks complaint → audio preprocessing → speech-to-text → text normalization → complaint category/intent classification → priority recommendation → complaint record creation → database storage → staff verification/assignment → status updates → user notification → resolution and closure.

3.4 Design Specifications

Component	**Proposed Design**
Voice Input	Browser/mobile microphone interface
Speech Recognition	Configurable ASR engine/API or approved open-source model
NLP	Tokenization/normalization plus embeddings or transformer-based representation
Classification	Supervised classifier or fine-tuned language model
Priority	Rules + model recommendation with human verification
Database	Relational database such as MySQL/PostgreSQL
Backend	Python-based API/service layer
Dashboard	Web interface for authorized staff
Authentication	Role-based access control
Notifications	Configurable email/SMS/in-app notification mechanism

3.5 Alternative Solutions and Trade-offs

Alternative	Advantages	Disadvantages	Decision
Manual classification	Simple, no model training	High staff effort, slower processing	Use only as fallback
Rule-based classification	Transparent and easy to implement	Limited handling of varied language	Useful baseline
Traditional ML classifier	Efficient and interpretable	Needs feature engineering	Suitable baseline
Transformer-based classifier	Better contextual understanding	Higher compute/model complexity	Preferred if resources permit
Cloud ASR	Strong speech recognition options	Internet dependency and data-privacy concerns	Use only with approved privacy controls
Local/open-source ASR	Greater control over data	May need more compute/tuning	Preferred where privacy is critical

3.6 Key Trade-off Analysis

- Accuracy versus response time: larger AI models may improve language understanding but increase computation and latency.
- Cloud versus local processing: cloud services can simplify deployment but require careful handling of complaint audio and transcripts.
- Automation versus human oversight: automation reduces repetitive work, while human verification reduces the impact of classification errors.
- Rich data collection versus privacy: storing audio can support auditing but creates additional privacy and storage requirements.
- General-purpose language models versus domain-specific models: general models provide flexibility, while domain-specific training can improve recognition of local complaint terminology.

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Mitigation
Sustainability	Computational and storage cost of audio/AI processing	Compress/delete audio according to retention policy; optimize models

Ethics	Voice and complaint information may contain personal details	Consent, minimization, anonymization and controlled retention
Safety	Urgent complaints may be incorrectly classified	Human verification and explicit escalation workflow
Security	Unauthorized access to complaint records	Authentication, role-based access, secure storage and audit logs
Bias	Speech/accent/language variation may affect recognition	Diverse evaluation data and error analysis
Reliability	ASR or network failure	Text fallback, retry mechanism and manual entry option

3.8 Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Speech recognition errors	High	Medium	High	Noise handling, clear prompts, confidence review, text correction	Medium
Wrong complaint category	Medium	High	High	Validation dataset, confidence threshold, human confirmation	Medium
Incorrect priority	Medium	High	High	Rule checks and staff approval for high-impact cases	Medium

Background noise	Medium	Medium	Medium	Audio preprocessing and re-record option	Low/Medium
Unauthorized data access	Low	High	High	RBAC, secure authentication and audit logs	Low/Medium
Database failure	Low	High	Medium	Backups and recovery procedures	Low
Model bias	Medium	Medium	Medium	Diverse test data and periodic evaluation	Low/Medium

3.9 Tools / Software / Equipment

Tool / Resource	**Purpose**
Python	Backend development, NLP and integration
Speech recognition model/API	Speech-to-text conversion
NLP / transformer library	Text preprocessing and classification
Pandas / NumPy	Dataset processing and analysis
Scikit-learn	Baseline machine-learning models and metrics
Database system	Complaint record storage
Flask/FastAPI/Django	Backend API layer
HTML/CSS/JavaScript or React	User and administrator interfaces
Git/GitHub	Version control and team collaboration
VS Code/Jupyter	Development and experimentation

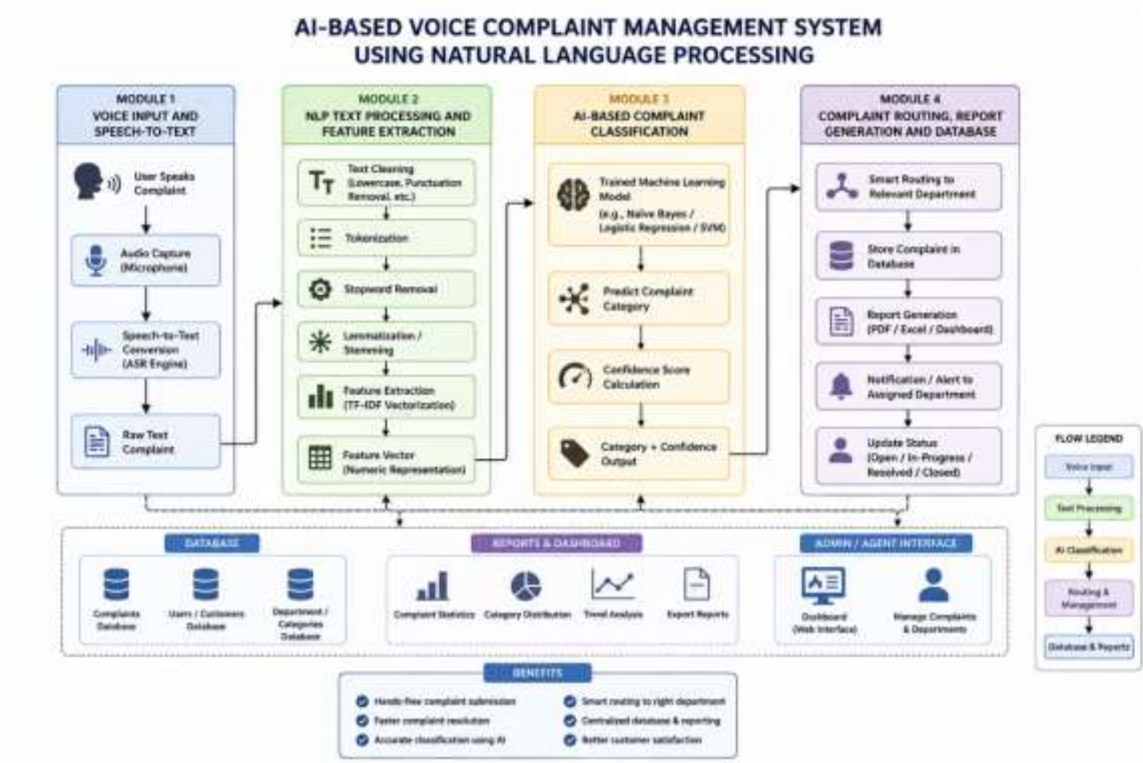

Fig. 3.1. Architecture Diagram

CHAPTER 4
IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project development can follow an iterative software-engineering approach consisting of requirement analysis, literature review, dataset preparation, voice-input development, speech recognition integration, NLP preprocessing, complaint classification, priority logic, database development, dashboard integration, testing, validation, and final documentation. Development should proceed in modules so that each component can be tested independently before system integration.

4.2 Hardware / Software / Modern Tools Used

The project is primarily software-based. A computer with a microphone, sufficient RAM and storage, and a stable development environment is required. If a local AI model is used, the computational requirement should be documented. If an external speech-recognition or AI API is used, the report should identify the service, version or model, data-handling constraints, and any applicable usage limits.

4.3 Dataset and Preprocessing

The dataset should contain representative complaint utterances mapped to the correct complaint category and, where applicable, priority level. The final report should state the source of the data, number of records, language(s), category distribution, train-validation-test split, and data-privacy/consent procedure. Text preprocessing may include normalization, removal of irrelevant symbols, handling of empty transcripts, spelling variation management, and conversion into model-ready representations.

If voice recordings are collected, the team should document the recording environment, microphone conditions, sample rate where relevant, language/accent coverage, consent procedure, and whether the audio is retained after transcription.

4.4 Voice Processing and Speech Recognition

The voice-processing module captures an audio segment after a user activates the recording control. The audio is passed to the selected speech-recognition component, which generates a transcript. The interface should provide feedback when recording begins, ends, or fails. A transcript-editing option can be provided so that users can correct recognition errors before final submission.

Speech recognition performance should be evaluated using an appropriate metric such as Word Error Rate (WER) or Character Error Rate (CER), depending on the language and evaluation setup. Results must be calculated from the actual test set used by the team.

4.5 NLP-Based Complaint Classification

The transcript is processed by the NLP layer to identify the complaint intent or category. A baseline model can be established using TF-IDF with a classical classifier, followed by comparison with a transformer-based or embedding-based approach if resources permit. The classification output should include a predicted category and, where supported, a confidence score. A confidence threshold can trigger manual verification when the system is uncertain.

For evaluation, the team should report accuracy, precision, recall, F1-score, and a confusion matrix where appropriate. Per-category results are useful because a high overall accuracy can conceal poor performance for less frequent complaint types.

4.6 Complaint Priority and Workflow

The priority module can use explicit urgency terms, category-specific rules, and/or model predictions to recommend Low, Medium, High, or Critical priority. Examples of signals may include words describing danger, service interruption, repeated failure, safety concerns, or time-sensitive impact. The recommendation should remain reviewable by authorized staff, especially for high-risk complaints. Each submitted complaint should receive a unique complaint ID. A typical lifecycle is Submitted → Verified → Assigned → In Progress → Resolved → Closed. The system should retain timestamps and relevant status history to support accountability and follow-up.

4.7 Dashboard and User Interface

The user interface should allow voice recording, transcript review, complaint submission, and complaint-ID display. The administrator dashboard should provide complaint search, category filtering, priority filtering, status filtering, assignment, detailed complaint viewing, and status updates. Summary indicators may show total complaints, unresolved complaints, high-priority complaints, and category-wise distribution. Where notifications are implemented, the system may provide acknowledgement after successful submission and status notifications when a complaint is assigned, updated, or resolved. Notification mechanisms should follow institutional privacy and communication policies.

4.8 Test Plan and Verification

Test ID	Test Case	Expected Result	Actual Result
T01	Record a clear voice complaint	Audio captured successfully	[Enter actual result]
T02	Convert speech to text	Readable transcript generated	[Enter actual result]
T03	Submit transcript	Unique complaint ID created	[Enter actual result]
T04	Classify known complaint	Correct category predicted	[Enter actual result]
T05	Ambiguous complaint	Low confidence/manual review triggered	[Enter actual result]
T06	Assign priority	Priority recommendation displayed	[Enter actual result]
T07	Update status	New status saved and visible	[Enter actual result]
T08	Search complaint ID	Correct record retrieved	[Enter actual result]
T09	Unauthorized dashboard access	Access denied	[Enter actual result]
T10	ASR/API failure	User receives clear error/fallback	[Enter actual result]

4.9 Results / Key Outcomes

The final submitted report should populate this section using the actual measured outcomes. The following table is provided as a reporting framework and must not be treated as fabricated project results.

Metric	Measured Value	Target / Reference	Observation
Speech recognition WER/CER	[Insert]	[Insert]	[Insert]

Complaint classification accuracy	[Insert]	[Insert]	[Insert]
Macro precision	[Insert]	[Insert]	[Insert]
Macro recall	[Insert]	[Insert]	[Insert]
Macro F1-score	[Insert]	[Insert]	[Insert]
Average complaint processing time	[Insert]	[Insert]	[Insert]
Successful complaint submission rate	[Insert]	[Insert]	[Insert]
Dashboard functional test pass rate	[Insert]	[Insert]	[Insert]

4.10 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence to Attach	Achievement
Voice complaint capture	Screenshots / demo / test record	To be evaluated
Speech-to-text conversion	Sample transcripts and ASR metrics	To be evaluated
Complaint classification	Model output, confusion matrix and metrics	To be evaluated
Priority recommendation	Rules/model outputs and test cases	To be evaluated
Complaint database and tracking	Database schema and workflow screenshots	To be evaluated
Management dashboard	Dashboard screenshots and functional tests	To be evaluated
Testing and validation	Test plan and actual results	To be evaluated
Usability and accessibility	User feedback / usability test evidence	To be evaluated

4.11 Strengths & Limitations

Strengths:

- Provides a voice-first mechanism for complaint registration.
- Combines speech recognition and NLP with a structured complaint workflow.
- Reduces repetitive manual categorization and data-entry effort.
- Provides centralized complaint tracking and dashboard-based monitoring.
- Can be extended to additional complaint categories and languages.
- Supports human verification rather than requiring fully autonomous decisions.

Limitations:

- Speech recognition performance may reduce in noisy environments or with unfamiliar accents.
- Classification quality depends strongly on the quality and diversity of the training/evaluation dataset.
- Priority recommendations may require human confirmation for sensitive cases.
- External AI/API services may introduce network dependency, cost, or data-governance concerns.
- The prototype may not represent production-scale performance unless load testing is performed.
- Language support may be limited unless multilingual data and models are evaluated.

4.12 Project Planning, Roles & Budget

Activity	Planned Period	Actual Period	Status
Requirement analysis	Project initiation	To be filled	To be filled
Literature review	Planning / analysis	To be filled	To be filled
Dataset preparation	Early implementation	To be filled	To be filled
Voice and ASR module	Implementation	To be filled	To be filled
NLP/classification module	Implementation	To be filled	To be filled
Database/workflow	Implementation	To be filled	To be filled
Dashboard integration	Integration	To be filled	To be filled
Testing and validation	Testing	To be filled	To be filled
Final documentation	Reporting	To be filled	To be filled

Student	Assigned Responsibility	Actual Contribution
Praharshini	Voice capture, requirements, speech-processing module	To be documented from actual work

Deeksha Sree	NLP preprocessing and complaint classification	To be documented from actual work
Hitaishi	Database, priority and complaint workflow	To be documented from actual work
Jeevitha	Dashboard, UI, notifications, testing/documentation	To be documented from actual work
Item	**Estimated Cost**	**Actual Cost**
Software libraries / open-source tools	Open-source / as applicable	To be filled
AI / speech API usage	To be estimated	To be filled
Cloud/database hosting	To be estimated	To be filled
Development computer / microphone	Existing / to be estimated	To be filled
Internet / other expenses	To be estimated	To be filled
TOTAL	To be calculated	To be calculated

CHAPTER 5
CONCLUSION & REFLECTION

5.1 Conclusion

The AI-Based Voice Complaint Management System addresses the difficulty of registering and managing complaints through a voice-enabled and AI-assisted workflow. The proposed solution connects voice capture, speech recognition, natural language processing, complaint classification, priority recommendation, structured storage, and dashboard-based monitoring. Instead of treating a complaint as an unstructured voice message, the system transforms it into a trackable record that can be assigned, updated, and resolved through a defined lifecycle.

The engineering value of the project lies in integrating multiple technologies while considering accuracy, usability, security, privacy, reliability, and human oversight. The final evaluation should establish how effectively the implemented system performs under the actual test conditions. The team should use measured results to discuss whether the objectives were fully or partially achieved and should clearly identify cases where the AI component produced errors.

Overall, the project demonstrates a practical application of artificial intelligence and data-driven software engineering to improve complaint registration and administrative workflow. With further validation and deployment-specific customization, the system can be extended to institutional, municipal, customer-service, campus, or other complaint-management contexts.

5.2 Future Work

- Support multiple Indian languages and code-mixed speech.
- Improve recognition under background noise and diverse acoustic conditions.
- Use larger and more representative complaint datasets for model training.
- Explore multilingual transformer models and domain-specific fine-tuning.
- Add sentiment, repeated-complaint detection, and duplicate-complaint identification.
- Introduce explainable classification and confidence-based human review.
- Integrate email/SMS/in-app notifications according to institutional policies.
- Add analytics for complaint trends, response time, department workload, and unresolved cases.

- Deploy the system on secure cloud or institutional infrastructure after proper security validation.
- Conduct larger-scale usability, fairness, accessibility, and performance testing.

5.3 Professional Learning & Individual Reflection

5.3.1 New Knowledge Acquired

- Speech recognition and audio-input processing.
- Natural language processing and text preprocessing.
- Supervised text classification and model evaluation.
- Database design and complaint workflow management.
- Web/API development and dashboard integration.
- Software testing, debugging, version control, and technical documentation.

5.3.2 Engineering & Professional Skills Developed

- Requirement analysis and system architecture design.
- Dataset preparation and data-quality assessment.
- Model selection and performance evaluation.
- Integration of AI components with conventional software systems.
- Privacy-aware and security-conscious system design.
- Team collaboration, task allocation, version control and technical communication.

5.3.3 Individual Reflection – Praharshini

During this project, I contributed to the development and validation of the AI-Based Voice Complaint Management System. My work involved the responsibilities assigned to my role and required coordination with the other team members. I gained practical experience in converting a real-world problem into a structured software solution, testing individual modules, identifying implementation issues, and improving the system based on observed results. I also learned the importance of documenting decisions, validating AI outputs, protecting user information, and communicating technical work within a team. The final reflection should be customized by each student to describe the exact modules implemented, specific challenges encountered, evidence of contribution, and the skills gained.

5.3.4 Individual Reflection – Deeksha sree

During this project, I contributed to the development and validation of the AI-Based Voice Complaint Management System. My work involved the responsibilities assigned to my role and required coordination with the other team members. I gained practical experience in converting a real-world problem into a structured software solution, testing individual modules, identifying implementation issues, and improving the system based on observed results. I also learned the importance of documenting decisions, validating AI outputs, protecting user information, and communicating technical work within a team. The final reflection should be customized by each student to describe the exact modules implemented, specific challenges encountered, evidence of contribution, and the skills gained.

5.3.5 Individual Reflection – Hitaishi

During this project, I contributed to the development and validation of the AI-Based Voice Complaint Management System. My work involved the responsibilities assigned to my role and required coordination with the other team members. I gained practical experience in converting a real-world problem into a structured software solution, testing individual modules, identifying implementation issues, and improving the system based on observed results. I also learned the importance of documenting decisions, validating AI outputs, protecting user information, and communicating technical work within a team. The final reflection should be customized by each student to describe the exact modules implemented, specific challenges encountered, evidence of contribution, and the skills gained.

5.3.6 Individual Reflection – Jeevitha

During this project, I contributed to the development and validation of the AI-Based Voice Complaint Management System. My work involved the responsibilities assigned to my role and required coordination with the other team members. I gained practical experience in converting a real-world problem into a structured software solution, testing individual modules, identifying implementation issues, and improving the system based on observed results. I also learned the importance of documenting decisions, validating AI outputs, protecting user information, and communicating technical work within a team. The final reflection should be customized by each student to describe the exact modules implemented, specific challenges encountered, evidence of contribution, and the skills gained.

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[6] McKinney, W., Python for Data Analysis, O’Reilly Media.

[7] Relevant official documentation for the selected speech-recognition technology, NLP framework, database, backend framework, and frontend framework used in the implementation.

[8] Actual journal/conference papers identified by the team during the documented literature-search process should be added here in the required institutional citation format.

APPENDICES

Appendix A – Sample Complaint Categories

Category	**Example Type of Complaint**
Infrastructure	Building, classroom, laboratory or facility-related issue
Maintenance	Electrical, plumbing, equipment or repair issue
Cleanliness	Waste, sanitation or housekeeping issue
Hostel	Hostel facility or service-related issue
Transport	Bus, route, timing or transport-service issue
Academic / Service	Academic administration or service-support issue
Other	Complaint not matching the predefined categories

Appendix B – Suggested Complaint Record Schema

Field	**Description**
Complaint ID	Unique identifier
User ID	Authorized reference to complainant
Timestamp	Date and time of submission
Audio Reference	Reference to audio if retained under policy
Transcript	Speech-to-text output
Category	AI-predicted / staff-confirmed category
Priority	AI-recommended / staff-confirmed priority
Assigned Department	Responsible handling unit
Status	Current workflow status
Resolution	Resolution details
Updated At	Latest status-update time

Appendix C – Evidence to Attach

- System architecture diagram.
- Voice-input interface screenshot.
- Speech-to-text sample outputs.
- Dataset/sample-data description.

- Classification model training/evaluation evidence.
- Confusion matrix and metric outputs.
- Complaint database schema.
- Administrator dashboard screenshots.
- Test records and verification results.
- Version-control evidence such as commits or repository history.
- Individual student contribution evidence.
- Presentation/demo evidence where permitted.

Appendix D – Ethics / Institutional Approval

The team should document whether institutional approval, participant consent, or a data-use permission process was required for collecting or processing voice complaints. If real user voice recordings are used, the final report should state the consent mechanism, purpose limitation, retention period, access controls, and deletion procedure. If synthetic, publicly available, or institutionally approved data are used instead, the source and usage conditions should be documented.

Appendix E – Individual Contribution Evidence

Student	**Evidence Type**	**Evidence Description**
Praharshini	Voice / preprocessing evidence	Code, recordings/test samples, ASR evaluation, screenshots
Deeksha Sree	NLP / classification evidence	Dataset preparation, model code, metrics and analysis
Hitaishi	Backend / database evidence	API, schema, workflow implementation and testing
Jeevitha	UI / dashboard evidence	Interface, dashboard, notification and usability testing

OUTPUTS

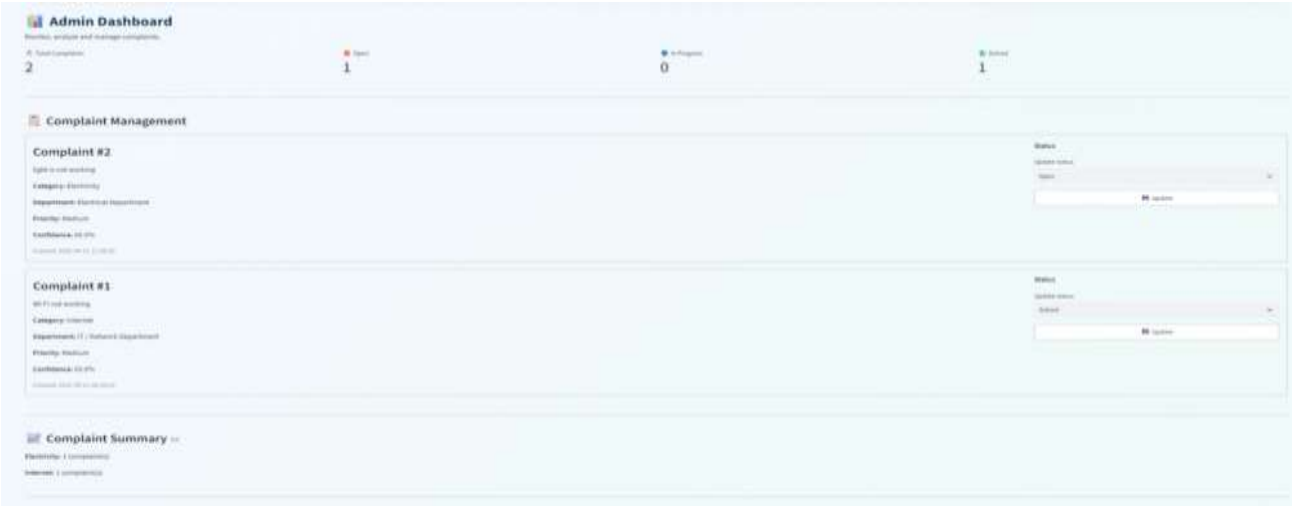

Figure 4.1 – Voice Complaint Recording and Speech-to-Text Output

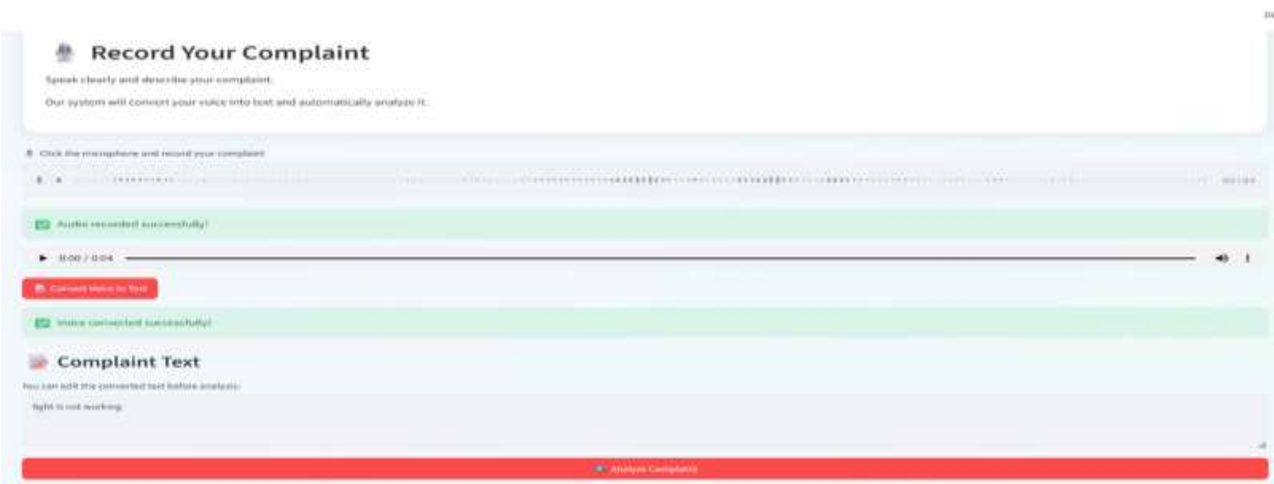

Figure 4.2 – AI Complaint Analysis and Smart Department Routing

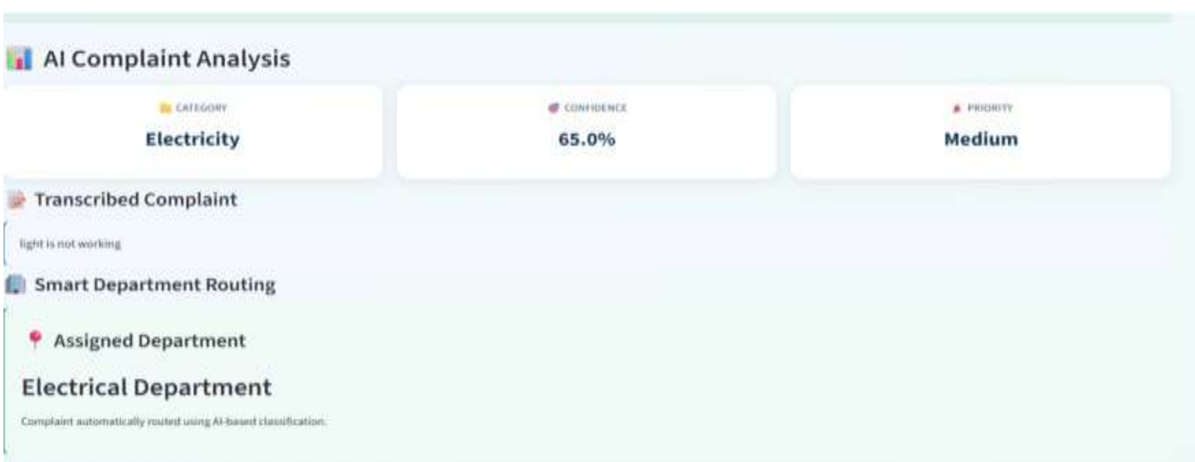

Figure 4.3 – Administrator Dashboard and Complaint Management

CAPSTONE PROJECT OUTCOME MAPPING SHEET

Outcome	Evidence	Location in Report
Complex engineering problem formulation	Problem statement and requirements	Ch. 1
Engineering design under constraints	Architecture, alternatives and trade-offs	Ch. 3
Written/oral technical communication	Whole report and viva	Whole report
Ethics and professional responsibility	Privacy, consent, security and responsible AI	Ch. 3
Teamwork and leadership	Roles, contribution statement and project plan	Ch. 4 + Appendix E
Experimentation, analysis and engineering judgment	Testing, metrics and comparison	Ch. 4
Independent acquisition of new knowledge	Professional learning and reflections	Ch. 5
Standards	Applicable software/data/security practices	Ch. 3
Sustainability and societal impact	Resource efficiency and accessibility	Ch. 3
Risk assessment and mitigation	Risk register	Ch. 3
Security considerations	Authentication, authorization and data protection	Ch. 3
Integrated/system-level engineering	End-to-end architecture and workflow	Ch. 3–4
Project planning and management	Timeline, roles and budget	Ch. 4
Individual contribution	Contribution statement and evidence	Front matter + Appendix E